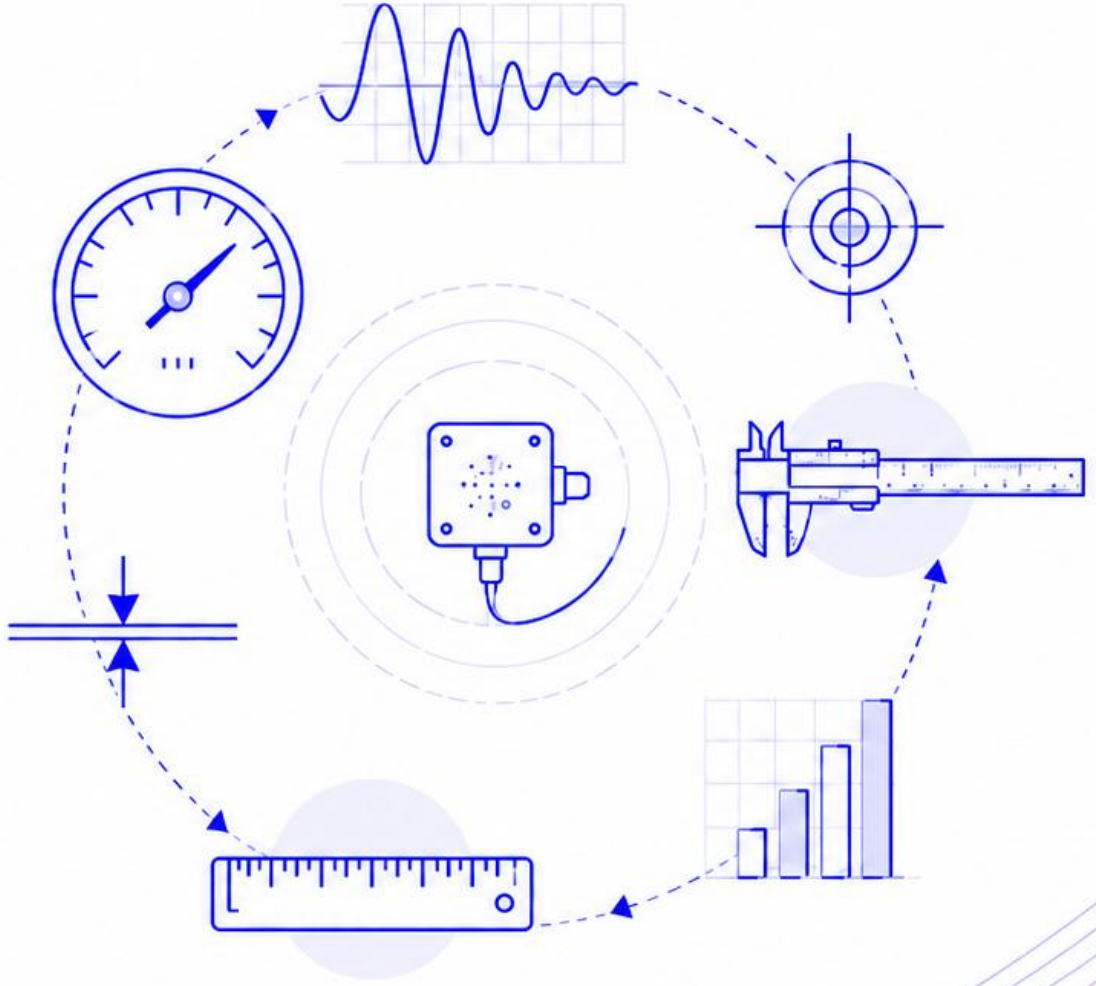
Measurement and Instrumentation for Energy Systems

Module 1: Measurement Fundamentals

by
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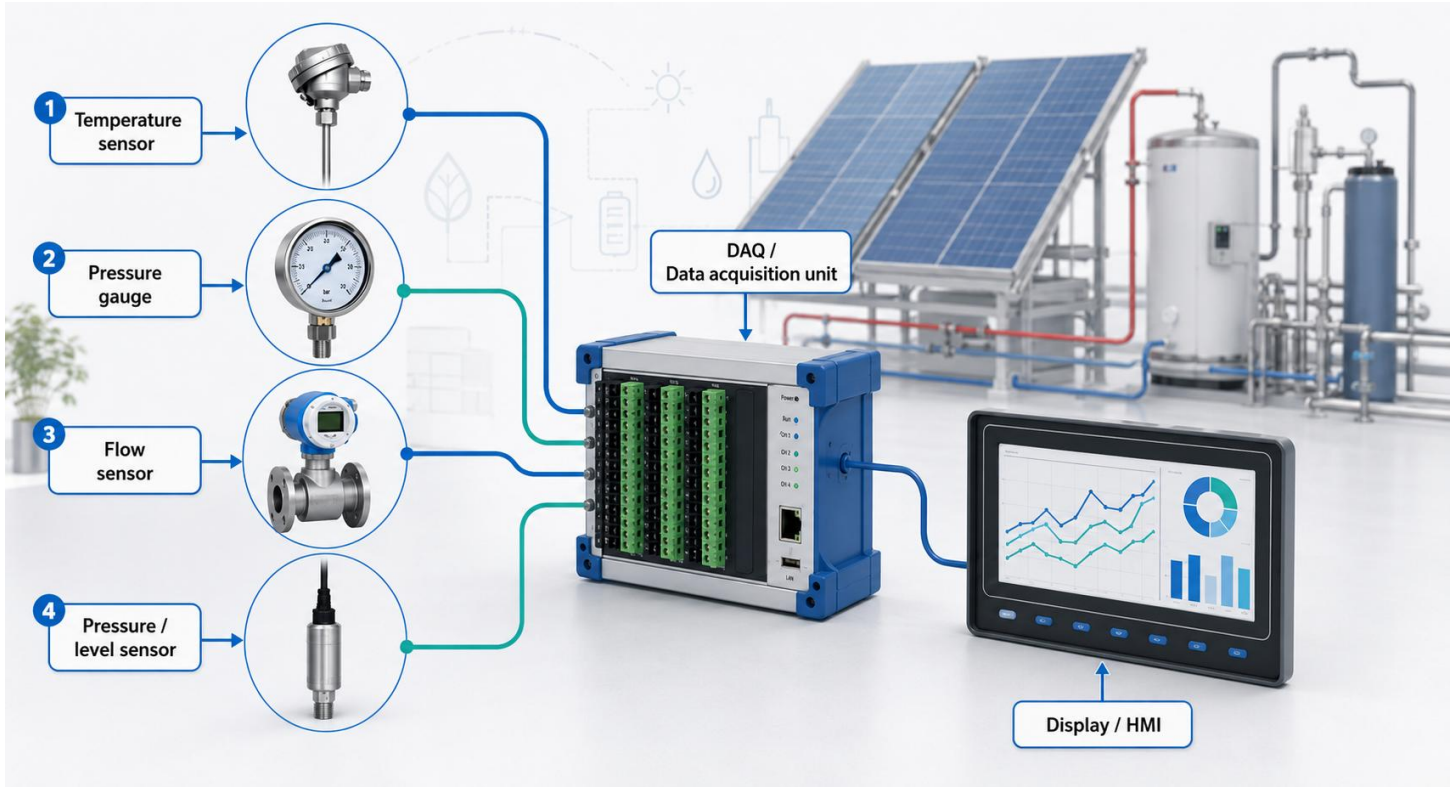
Contents

- Fundamental and derived quantities
- Static and dynamic concepts
- Sensitivity
- Stability
- Resolution
- Accuracy
- Precision
- Calibration
- Errors



What is Measurement?

- Measurement is the process of determining the magnitude or value of a physical quantity.
- The quantity being measured is called the measurand.
- A measurement system converts information about a physical quantity into a usable output.
- The measured value should represent the true value as closely as possible.



Typical energy-system measurement and data acquisition chain

Process /Energy system → **Sensor** → **Signal conditioning** → **Display /DAQ** → **Measured value**

References: Morris & Langari; Bentley; Venkateshan.

Fundamental Quantities

- Fundamental quantities are independent base quantities in a system of units.
- They are not obtained by combining other physical quantities.
- They are the building blocks used to define other engineering quantities.

Quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

References: Morris & Langari, Ch. 1; Bentley, Ch. 1; Venkateshan nomenclature.

Derived Quantities

- Derived quantities are obtained by mathematically combining fundamental quantities.
- Most quantities measured in energy systems are derived quantities.
- Their units are also derived from SI base units.

Derived quantity	Relation	SI unit
Velocity	Length / Time	m/s
Acceleration	Velocity / Time	m/s^2
Density	Mass / Volume	kg/m^3
Force	Mass × Acceleration	N
Pressure	Force / Area	Pa
Flow rate	Volume / Time	m^3/s
Power	Energy / Time	W

Pressure = Force / Area
$Pa = N/m^2$

Power = Energy / Time
W = J/s

References: Morris & Langari, Ch. 1; Venkateshan, nomenclature.

Static Measurement Concept

- A measurement is treated as static when the measurand is constant or changes very slowly.
- The instrument has enough time to settle and follow the input.
- Static characteristics describe the steady input-output relation.
- Examples: Room temperature, Tank level at steady condition, Constant DC voltage.

True value 50 °C → **Thermometer settled** → **Reading 50 °C**

Main concern: Is the final reading correct?

Input ≈ constant
dX/dt ≈ 0

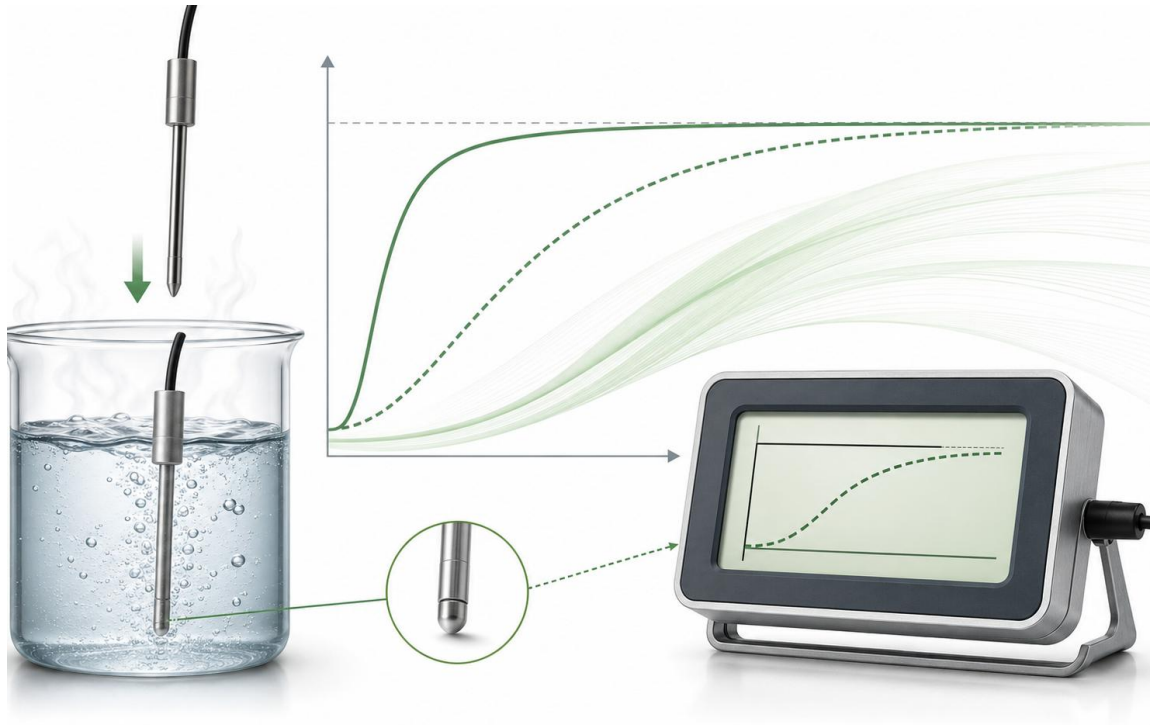
Output after settling
O = f(I)

- X = quantity being measured, called the measurand
- t = time
- $\frac{dX}{dt}$ = rate at which the measurand changes with time
- I = instrument input
- O = instrument output

References: Bentley, Ch. 2; Morris & Langari, Ch. 2.

Dynamic Measurement Concept

- A measurement is dynamic when the measurand changes significantly with time.
- The instrument may not follow the input instantly.
- There can be a difference between true instantaneous value and indicated value.
- Examples: Furnace start-up temperature, Vibration, Pressure pulsation, Flow fluctuations.



Temperature sensor response to a sudden change

Dynamic error = True instantaneous value - Indicated value

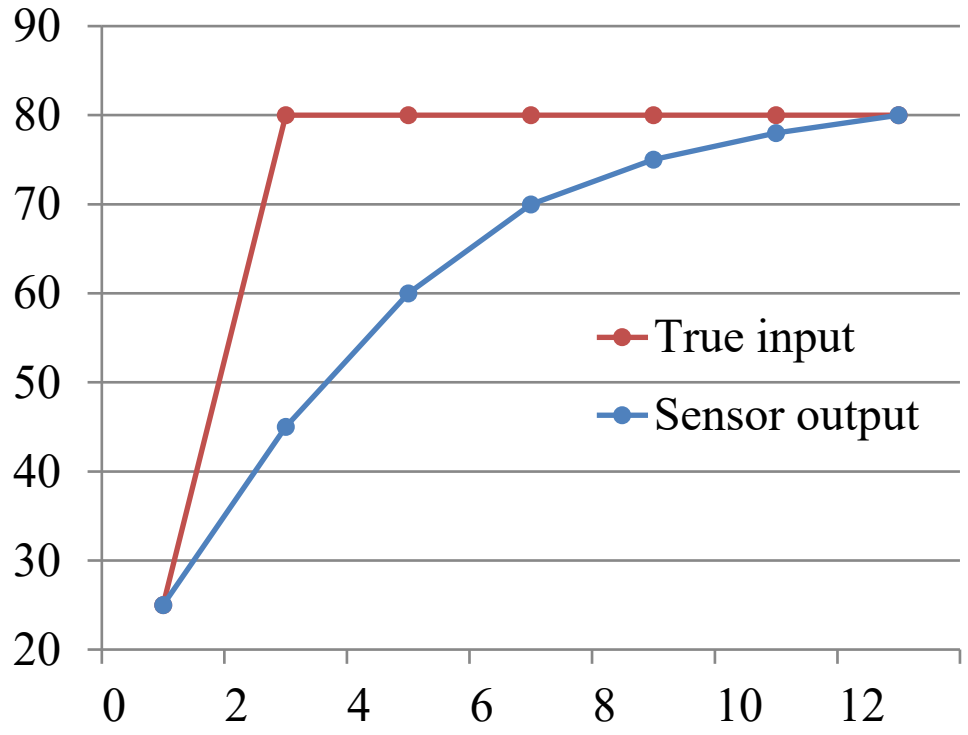
Main concern: How fast does the instrument follow the changing input?

References: Bentley, Ch. 4; Morris & Langari, Ch. 2.

Static vs Dynamic Measurement: Same Sensor, Different Situation

- The quantity may be the same, but the measurement situation can be different.
- A thermometer can be suitable for static measurement but too slow for dynamic measurement.
- Instrument selection must consider both final accuracy and response speed.

Situation	Static measurement	Dynamic measurement
Temperature	Room after settling	Furnace heating transient
Pressure	Storage tank pressure	Engine cylinder pressure
Flow rate	Steady pipe flow	Pulsating pump discharge
Main question	Is the final value correct?	Can the instrument follow changes?



Actual input changes faster than the sensor output

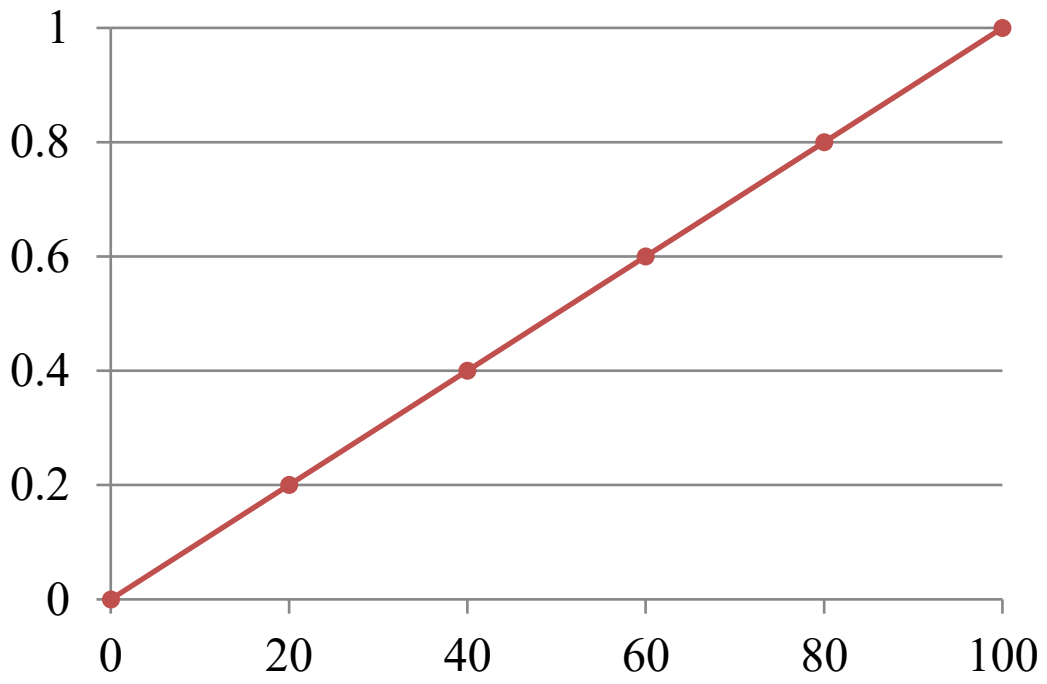
References: Bentley, dynamic characteristics; Morris & Langari, instrument response.

Sensitivity

- Sensitivity tells how much output changes for a given input change.
- It is the slope of the instrument input-output relation.
- High sensitivity helps to produce a larger output signal for small input changes.
- High sensitivity alone does not guarantee high accuracy.

Sensitivity = ΔOutput / ΔInput

For nonlinear case: S = dO/dI



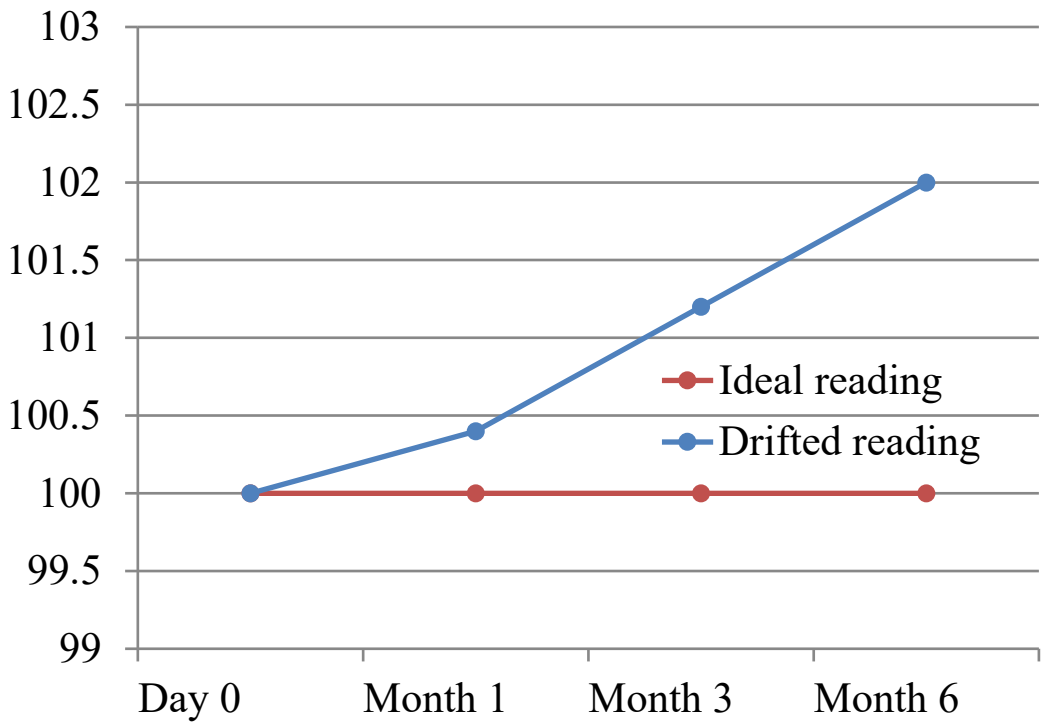
Example:
0-100 °C gives 0-1 V
Sensitivity = 1/100 = 0.01 V/°C = 10 mV/°C

References: Fraden, sensor characteristics; Morris & Langari, static characteristics; Bentley, Ch. 2.

Stability and Drift

- Stability is the ability of an instrument to maintain its characteristics over time.
- If the input is unchanged but the output changes slowly, the instrument is drifting.
- Drift may occur due to ageing, temperature exposure, humidity, wear or component degradation.
- Stable instruments require less frequent recalibration.

Same input over time → ideally same output



Example: A sensor kept at a true 100 °C reading gradually shifts to 102 °C after months.

References: Fraden, stability and reliability; Bentley, static characteristics.

Resolution

- Resolution is the smallest change in input that produces a detectable output or display change.
- For digital instruments, resolution is often governed by ADC bits and display steps.
- Good resolution does not automatically mean good accuracy.

Instrument	Displayed values	Resolution
Thermometer A	25, 26, 27 °C	1 °C
Thermometer B	25.0, 25.1, 25.2 °C	0.1 °C
Thermometer C	25.00, 25.01 °C	0.01 °C

Digital resolution ≈ Full-scale range / 2^n

Example: A 12-bit system over 0-100 °C has ideal digital step ≈ 100/4096 = 0.024 °C.

References: Fraden, resolution; Morris & Langari, static characteristics.

Sensitivity and Resolution: Relationship

- Sensitivity is output change for a unit input change
- Resolution is smallest detectable input change.
- For both system A and B, if electronics used and noise level are same, higher sensitivity usually improves the resolution.
- But better electronics can give better resolution even with lower sensor sensitivity.

System	Sensitivity	Minimum detectable output	Resolution
A	10 mV/kg	1 mV	0.1 kg
B	1 mV/kg	0.001 mV	0.001 kg

System B has lower sensitivity, but better resolution because its electronics can detect much smaller output changes.

References: Morris & Langari; Bentley; Fraden.

Accuracy

- Accuracy describes closeness of a measured value to the true or reference value.
- A perfectly accurate instrument is an ideal case.
- Accuracy is judged by measurement error relative to a known standard.
- Systematic error directly affects accuracy.



Closer reading → better accuracy

Error = Measured value - True value

Example:
True temperature = 100 °C
Measured temperature = 98 °C
Error = -2 °C

- Accuracy is usually specified as ± value or ±% of full scale.
- Calibration compares the instrument with a reference standard to evaluate its accuracy.
- Accuracy is not the same as resolution.

References: Bentley, Ch. 1 and Ch. 3; Morris & Langari, Ch. 2.

Precision

- Precision describes the closeness of repeated measurements to each other.
- It is associated with repeatability and scatter of readings.
- A precise instrument gives nearly the same reading on repeated trials.
- Precision does not require the readings to be close to the true value.

Repeated readings for true 10 kg	Precision
9.82, 9.81, 9.82, 9.81 kg	High
9.5, 10.4, 9.8, 10.3 kg	Low

The first set is precise because the values are close to each other; however it may still be inaccurate if the true value is 10 kg.

Precision = consistency of repeated readings

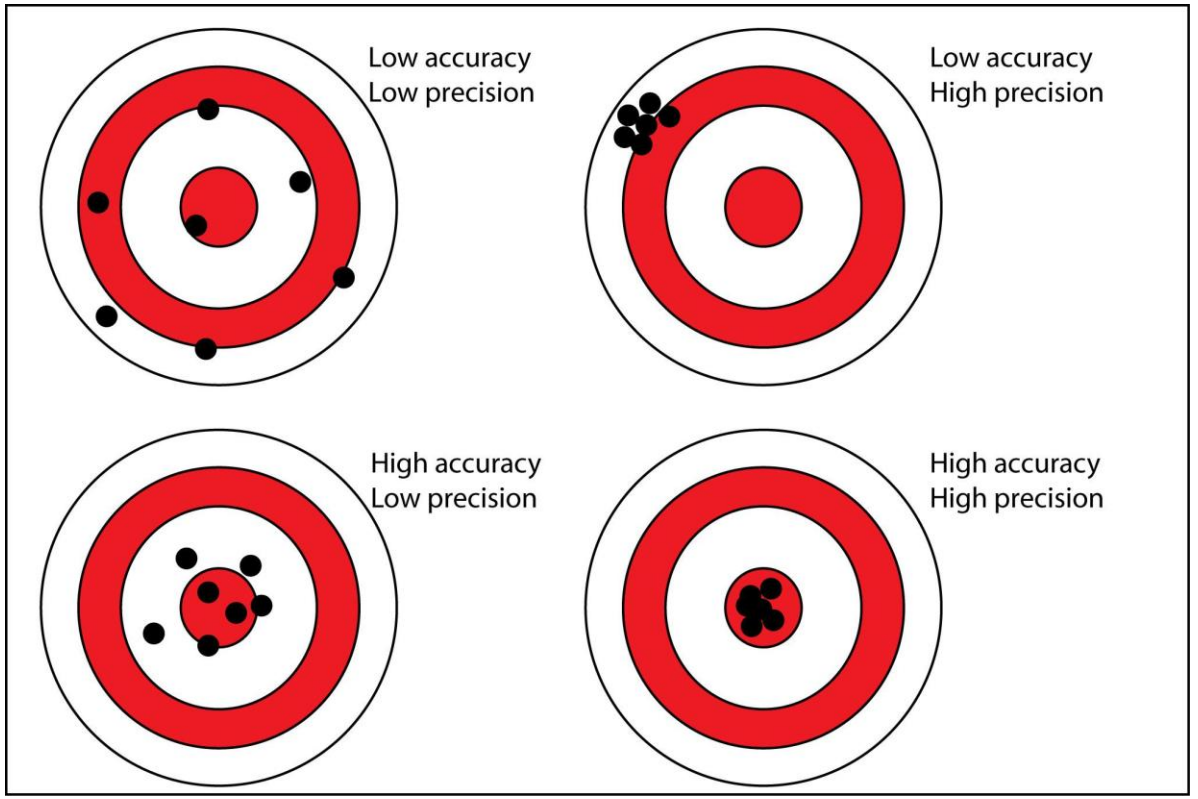
High precision → small scatter

References: Morris & Langari, precision/repeatability; Bentley, statistical characteristics.

Accuracy vs Precision

- Accuracy concerns closeness to the true value.
- Precision concerns closeness among repeated readings.
- An instrument can be precise but inaccurate if it has a systematic bias.
- Random error mainly affects precision; systematic error mainly affects accuracy.

Term	Main question
Accuracy	Is the average reading close to the true value?
Precision	Are repeated readings close to one another?



Target analogy for accuracy and precision

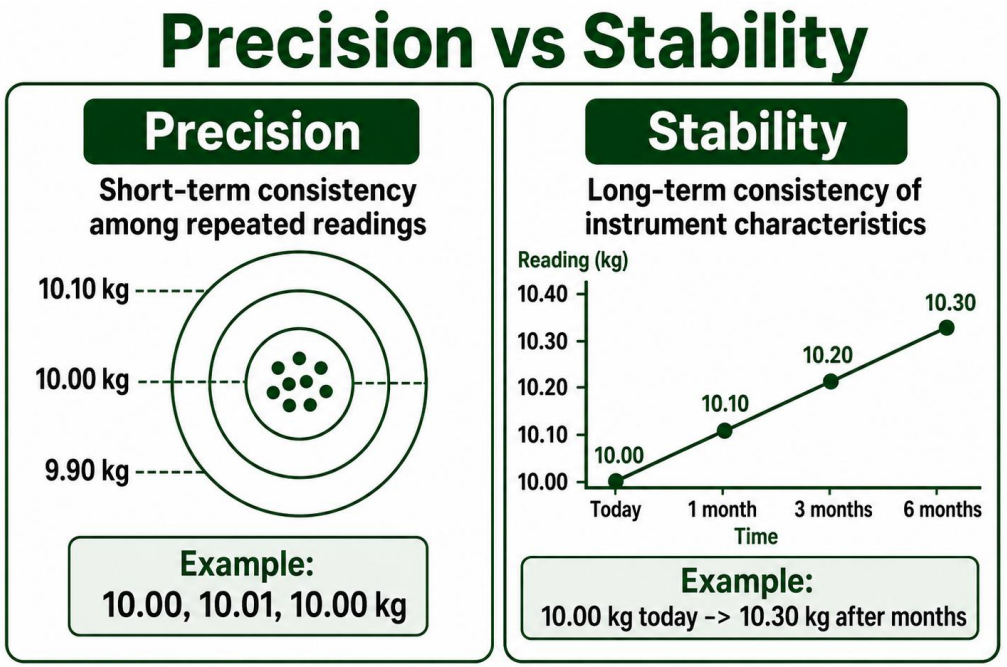
References: Morris & Langari, Ch. 2 and Ch. 3; Bentley, Ch. 2.

Precision vs Stability

- Precision is short-term consistency among repeated readings.
- Stability is long-term consistency of instrument characteristics.
- A system can be precise today but unstable over months due to drift.
- Stability is checked by repeated calibration over time.

Feature	Precision	Stability
Concern	Scatter in repeated readings	Drift with time
Time scale	Short interval	Long interval
Example	10.00, 10.01, 10.00 kg	10.00 kg today → 10.30 kg after months

Precision: repeatability now
Stability: repeatability of performance over time



References: Fraden, stability; Morris & Langari, precision/repeatability.

Errors in Measurement

Error (Experimental Error)

The difference between an instrument's measured value and the known or true value of the quantity being measured.

Uncertainty

The estimated boundary or range within which the actual error is expected to lie, commonly stated with specific confidence levels (i.e. 50%, 80%, 90%, 95%, 98%, 99%, and 99.9%).

Types of Errors in Measurement:

Non-Random (Systematic) Errors

Errors that cause repeated measurements to be consistently high or low by approximately the same amount, which can often be corrected through calibration or theoretical analysis.

- Uncorrected Environmental & Physical Loading
- Improper Calibration or Standard Offset
- Instrument Hysteresis & Mechanical Deficiencies
- Construction & Design Flaws

Random Errors

Unpredictable variations caused by electrical noise, friction, or environmental changes that can be analyzed using statistical methods.

- Observer Fluctuations
- Random Electronic Noise
- Mechanical & Dynamic Fluctuations
- Uncontrolled Environmental Fluctuations

Types of Data Sampling

Single-Sample Data

Data collected from a single setup or instrument where systematic errors cannot be identified merely by repeating the measurement.

Multisample Data

Data collected using multiple independent instruments or setups to ensure reliability through statistical evaluation.

Statistical Measures

- To quantify the spread of random errors
- Estimate the true value and precision of a dataset
- Determine the confidence intervals
- Uncertainties associated with experimental measurements.

Geometric Mean (x_g)

The nth root of the product of "n" observations, utilized for evaluating phenomena exhibiting growth rates proportional to their instantaneous magnitude.

Arithmetic Mean (x_m)

The primary estimator of the true value of an observed physical variable, defined as the sum of all individual readings divided by the total number of observations n:

$$x_m = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Where x_i= Value of Parameter x in the ith measurement

$$x_g = [x_1 \cdot x_2 \cdot x_3 \cdots\cdots x_n]^{1/n}$$

Types of Data Sampling

Harmonic Mean (x_m)

The harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocals of a given set of non-zero positive observations:

$$H = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

Mode

The mode is the measure of central tendency representing the value that occurs with the highest frequency in a data set. The modal value is estimated within the modal class (the class interval with the highest frequency) using:

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \cdot h$$

Median

The positional parameter representing the middle value that divides a sorted set of data points into two equal halves.

Odd n: The median is the single middle value located at position $\frac{n+1}{2}$

$$\text{Median} = x_{\left(\frac{n+1}{2}\right)}$$

Even n: The median is the arithmetic mean of the two middle values located at positions $\frac{n}{2}$ and $\frac{n+1}{2}$

$$\text{Median} = \frac{x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n+1}{2}\right)}}{2}$$

where: L = lower class limit of the modal class, h = class interval width, f_1 = frequency of the modal class, f_0 = frequency of the preceding class, f_2 = frequency of the succeeding class

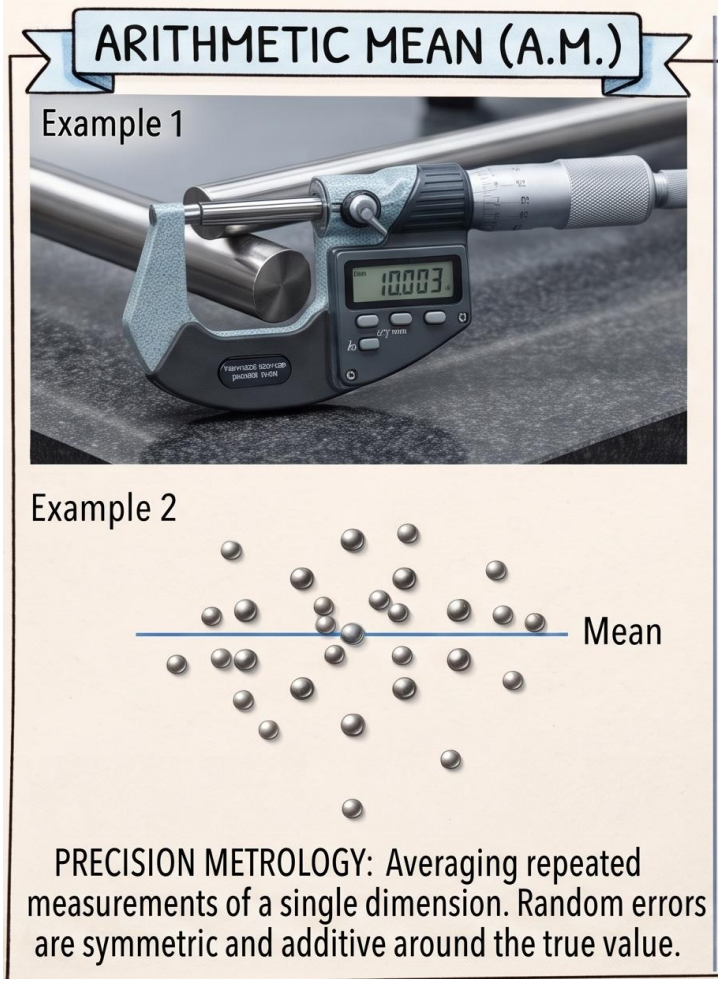
Arithmetic Mean

Arithmetic Mean (Repeated Physical Measurements):

- When measuring a physical dimension like the length of a metal rod repeatedly, errors arise from tiny, independent fluctuations (like slight hand tremors, minor temperature variations, or small reading inconsistencies).

- These errors are additive - they randomly push your measurement a little too high or a little too low with equal probability.

- The arithmetic mean naturally cancels out these symmetric, random errors because the positive and negative deviations average out to zero, giving you the best estimate of the true length/value.



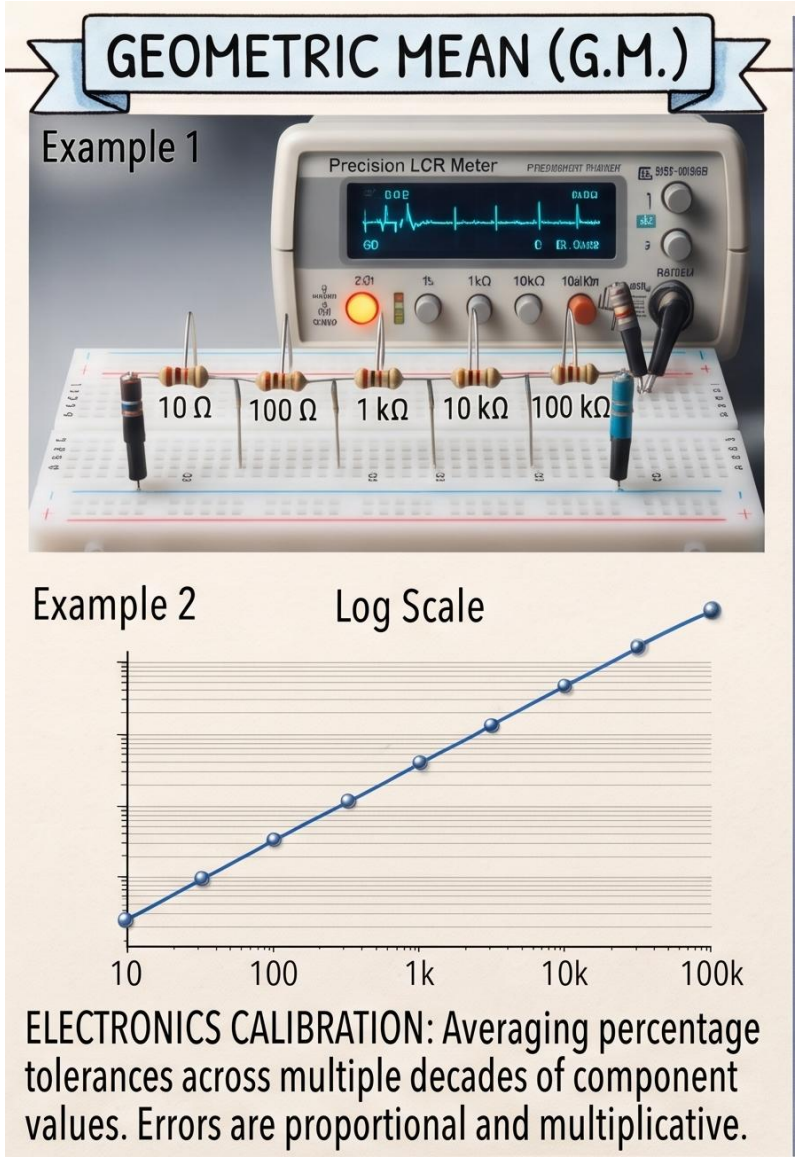
Geometric Mean

Geometric Mean (Resistor Calibration & Proportional Errors)

- In electronics, component tolerances and calibration errors are rarely fixed absolute amounts; instead, they scale proportionally (e.g., a 5% error).

- If you are testing resistors across vastly different orders of magnitude (say, 10 Ω up to 1,000,000 Ω, standard arithmetic averaging would let the massive absolute errors of the million-ohm resistors completely dominate the calculation.

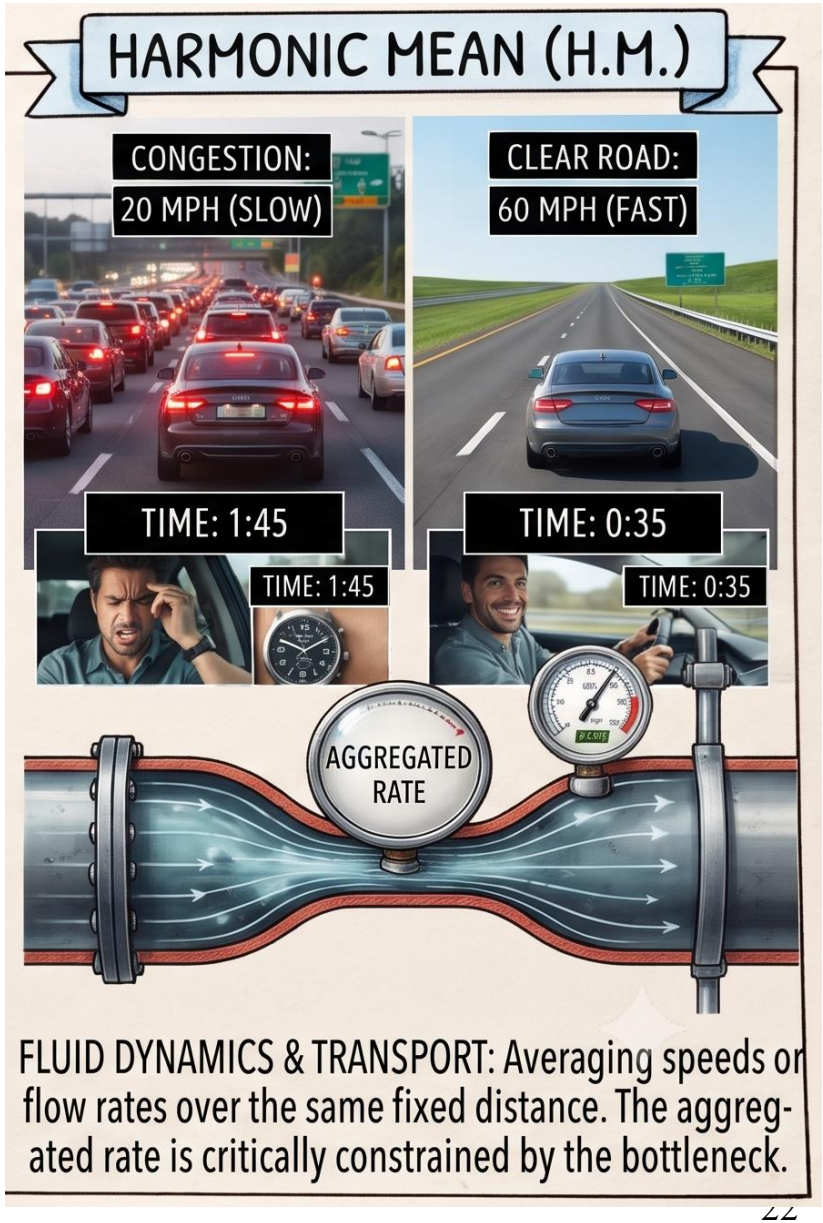
- The geometric mean operates on a multiplicative or logarithmic scale, ensuring that percentage-based and proportional errors are weighted evenly regardless of the component's base value.



Harmonic Mean

Harmonic Mean

- The Ultimate Real-World Example: Driving Speed
- Imagine you drive a fixed distance of 60 miles to a neighboring city. On the way there, traffic is light, and you drive at 60 mph (taking 1 hour).
- On the way back, you hit heavy traffic and drive at 20 mph (taking 3 hours).What is your actual average speed for the whole round trip? Your instinct might be to use the normal arithmetic mean: $\frac{60+20}{2} = 40$.
- That answer is completely wrong.
- To find your real average speed, you have to look at the total distance divided by the total time:
- Total distance = 60 miles (there) + 60 miles (back) = 120 miles.



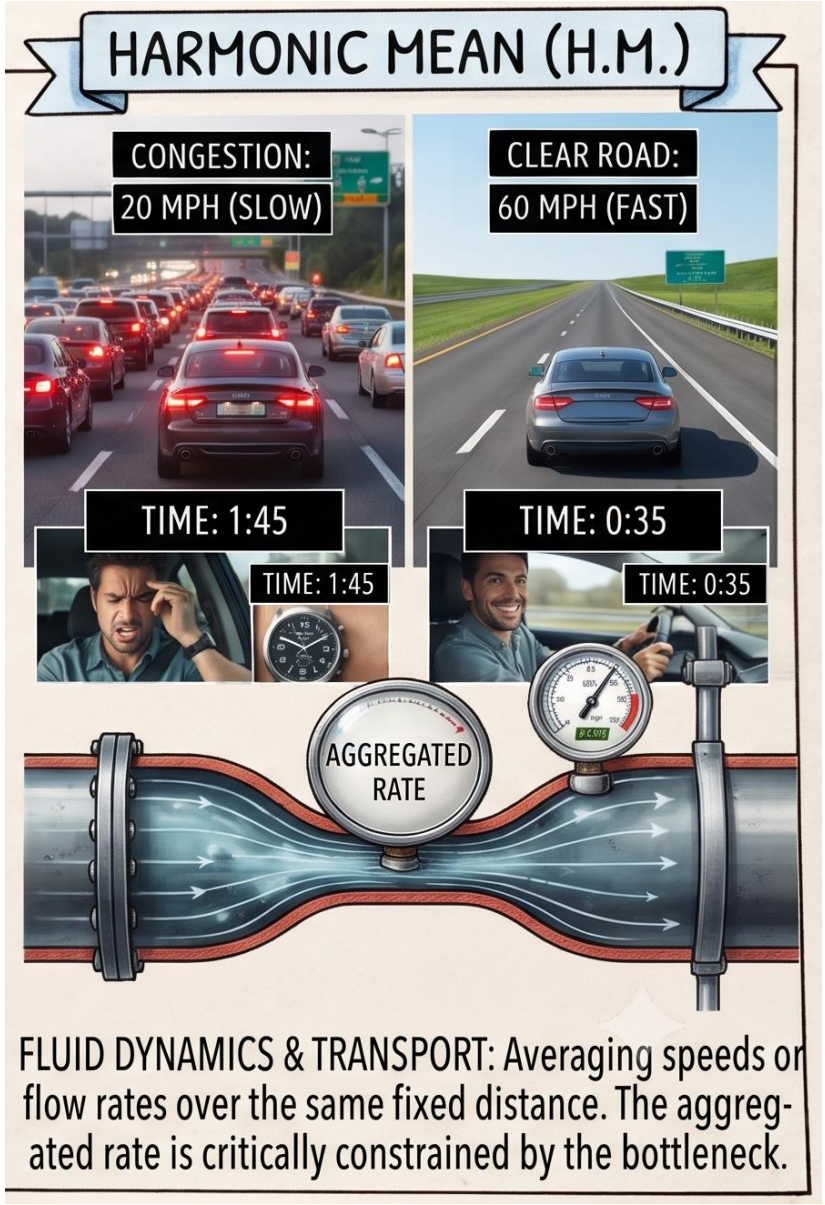
Harmonic Mean

Total time = 1 hour (there) + 3 hours (back) = 4 hours.

Real Average Speed = 120/4=30 mph.

How the Harmonic Mean solves this:

If you plug your two speeds (60 and 20) into the harmonic mean formula, it automatically handles the "time spent" imbalance and gives you the exact correct answer of 30 mph. It doesn't get tricked by the fact that you spent more time driving at the slower speed.



Mode

- Imagine you are using a digital kitchen scale and place a precise **100-gram calibration weight** on it 10 separate times.
- Because of tiny electrical noise or digital rounding, the scale doesn't show "100.0 g" every single time. Instead, it gives you this spread of readings:
  - **99.9 g** (appears 2 times)
  - **100.0 g** (appears 6 times)
  - **100.1 g** (appears 2 times)
- The **mode** is **100.0 g** because it is the reading that shows up most frequently (6 out of 10 times).



Why It Matters in Measurement

Finding the True State: In digital measurement instruments, the mode tells that the exact output code or value the device settles on most reliably, stripping away random, transient noise.

Catching Calibration Drift: If the test is performed one day later tomorrow and the mode suddenly shifts to **100.3 g**, the mode instantly flags a systematic calibration error. It indicates that the instrument's baseline has drifted, whereas a standard average might occasionally mask the shift if combined with other random fluctuations.

Examples of Arithmetic Mean and Geometric Mean

Arithmetic Mean in Measurement

- Repeated Sensor Calibration Readings: Averaging multiple test readings together to smooth out random errors and find the true value.
- Dimensional Metrology: Measuring an object several times with a tool and averaging the results to check its exact size.
- Analytical Mass Determination: Weighing an item multiple times on a scale to cancel out tiny environmental disturbances.
- Spectrophotometric Absorbance Trials: Averaging multiple light absorption tests to find a chemical solution's reliable concentration.
- Acoustic Noise Level Logging: Averaging sound level measurements across a space to find the true background noise level.

Geometric Mean in Measurement

- Particle Size Distribution: Finding the average size of soil or powder particles that vary wildly across massive scales.
- Microbial Colony Enumeration: Averaging bacteria counts that grow exponentially rather than linearly.
- Optical Attenuation and Transmittance: Averaging how much light passes through multiple stacked layers of filters.
- Acoustic and Vibrational Frequency Ratios: Averaging pitch or frequency scaling steps that multiply instead of add.
- Relative Growth and Scaling Factors: Averaging proportional changes that compound multiplicatively.

Examples of Harmonic Mean and Mode

Harmonic Mean in Measurement

Fluid Flow Velocity Over Equal Distances: Finding the true average speed when moving across equal distances at different rates.

Parallel Circuit Resistance and Impedance: Calculating the total resistance of electrical components wired in parallel.

Resource Efficiency Rates: Finding the true average fuel economy when driving identical distances at varying speeds.

Volumetric Pumping and Discharge Rates: Calculating the effective rate when systems process fixed volumes of liquid.

Optical Focal Lengths and Conjugate Distances: Combining lens distances to calculate overall optical properties.

Mode in Measurement

Prevailing Wind Direction: Determining the most frequent wind direction recorded by a weather station over a given time period.

Manufacturing Quality Control: Identifying the most commonly produced part size or defect category on an assembly line to track a machine's primary tendency.

Retail Size Inventory Selection: Finding the most frequently measured customer size (such as shoe or ring dimensions) to stock the highest-demand products.

Digital Sensor State Logging: Finding the most frequent operational state (such as ON/OFF or specific error codes) logged by an electronic monitor over time.

Traffic Flow Peak Counting: Tracking the most common vehicle count recorded during standard time intervals to identify typical congestion levels.

Problem

Q) During an experimental flow-loop measurement in a fluid mechanics laboratory, local fluid velocities (u, in m/s) were recorded by a Pitot-static tube across thirteen cross-sectional subsections of a circular pipe. Due to turbulent fluctuations and randomized spatial sampling along the cross-section, the data acquisition system logged the readings in the following unsorted sequence:

4, 5, 8, 10, 6, 12, 6, 8, 4, 8, 6, 10, 6 m/s

To completely characterize the flow velocity distribution and interpret the related transport characteristics, calculate the following five statistical parameters for this dataset:

a) Mode

b) Median

c) Arithmetic Mean

d) Geometric Mean

e) Harmonic Mean

f) Standard deviation

g) Variance

Solution

a) Mode

The mode is the value that occurs most frequently.
The highest frequency is 4, corresponding to 6 m/s.
So Mode = 6 m/s

b) Median

To calculate the median, the data must first be arranged in order.
So arrange the data in ascending order:
4, 4, 5, 6, 6, 6, 6, 8, 8, 8, 10, 10, 12
For n=13, the median position is: : (n+1)/2 = (13+1)/2 = 7th
The 7th value in the ordered data is: Median = 6 m/s

c) Arithmetic Mean

$$x_m = \frac{\sum x_i}{n}$$

$$= \frac{4+5+8+10+6+12+6+8+4+8+6+10+6}{5}$$

$$= 93/13$$
$$= 7.15 \text{ m/s}$$

d) Geometric Mean

The geometric mean(GM) is:
$GM = (x_1 x_2 x_3 \ldots x_n)^{1/n}$
$GM = (4\times5\times8\times10\times6\times12\times6\times8\times4\times8\times6\times10\times6)^{1/13}$
$GM = (63{,}700{,}992{,}000)^{1/13}$
GM = 6.78 m/s

e) Harmonic Mean

$$H = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

$$= \frac{13}{\frac{1}{4}+\frac{1}{5}+\frac{1}{8}+\frac{1}{10}+\frac{1}{6}+\frac{1}{12}+\frac{1}{6}+\frac{1}{8}+\frac{1}{4}+\frac{1}{8}+\frac{1}{6}+\frac{1}{10}+\frac{1}{6}}$$

= 13/2.025
= 6.42 m/s

Solution

f) Standard deviation

$$\sigma = \left[\frac{\sum_{i=1}^{n}(x_i - x_m)^2}{n-1}\right]^{1/2}$$

First calculate:

$\sum x_i^2$= 16+25+64+100+36+144+36+64+16+64+36+100+36

= 737

$$\sigma^2 = \frac{\sum x_i^2}{n} - x_m^2$$

$$\sigma^2 = \frac{737}{13} - (7.15)^2$$

$$\sigma^2 = 56.692 - 51.178$$

$$\sigma^2 = 5.515$$

$$\sigma = \sqrt{5.515}$$

$$\sigma = 2.35\ m/s$$

g) Variance

$\sigma^2 = 5.515$

Types of Data Sampling

During an experimental test on a solar thermal receiver plate, surface temperature readings are logged across multiple thermocouple zones. Calculate the mode(s) for the following continuous frequency distribution of recorded temperatures (°C), which features two non-adjacent class intervals sharing the highest frequency:

Temperature Interval (°C)	Frequency (f - Number of Readings)
300 – 320	3
320 – 340	10
340 – 360	6
360 – 380	10
380 – 400	4

When a grouped frequency distribution contains two or more modal classes with the same maximum frequency, the distribution is bimodal. Each modal class is evaluated independently using the grouped mode formula:

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \cdot h$$

where: L = lower class limit of the modal class, h = class interval width,

f_1 = frequency of the modal class, f_0 = frequency of the preceding class,

f_2 = frequency of the succeeding class

Types of Data Sampling

First Modal Class: 320 – 340 °C

- Lower limit (L) = 320
- Interval width (h) = 20
- Modal frequency (f1) = 10
- Preceding frequency (f0) = 3 (frequency of interval 300 – 320)
- Succeeding frequency (f2) = 6 (frequency of interval 340 – 360)

Mode1 = 320 + [(10 - 3) / (2(10) - 3 - 6)] × 20

Mode1 = 320 + [7 / (20 - 9)] × 20

Mode1 = 320 + (7 / 11) × 20 = 320 + 12.73 = 332.73 °C

Second Modal Class: 360 – 380 ° C

- Lower limit (L) = 360
- Interval width (h) = 20
- Modal frequency (f1) = 10
- Preceding frequency (f0) = 6 (frequency of interval 340 – 360)
- Succeeding frequency (f2) = 4 (frequency of interval 380 – 400)

Mode2 = 360 + [(10 - 6) / (2(10) - 6 - 4)] × 20

Mode2 = 360 + [4 / (20 - 10)] × 20

Mode2 = 360 + (4 / 10) × 20 = 360 + 8.00 = 368.00 °C

The temperature distribution of the receiver plate is bimodal, revealing two distinct most-frequent operating temperature states:

- First Mode: 332.73°C
- Second Mode: 368.00 °C

Calculation of Mode

During an experiment on flow through a pipe, the **flow velocity** was measured at 60 different time intervals. The observations were grouped as follows:

Flow velocity, *V*(m/s)	**Number of observations**
0.0–0.5	3
0.5–1.0	7
1.0–1.5	**15**
1.5–2.0	**15**
2.0–2.5	10
2.5–3.0	7
3.0–3.5	3
Total	**60**

Calculate the mode of the flow velocity using the grouped-data mode formula, considering that **1.0–1.5 m/s and 1.5–2.0 m/s are two adjacent modal classes having the same maximum frequency**.

Calculation of Mode

$$\text{Mode} = \text{L} + \left(\frac{\text{f}_1 - \text{f}_0}{2\text{f}_1 - \text{f}_0 - \text{f}_2}\right) \cdot \text{h}$$

where: L = lower class limit of the modal class, h = class interval width, f_1 = frequency of the modal class, f_0 = frequency of the preceding class, f_2 = frequency of the succeeding class

Case 1: Taking 1.0–1.5 m/s as the modal class

$$L = 1.0, f_1 = 15, f_0 = 7, f_2 = 15, h = 0.5$$

Therefore,

$$\text{Mode} = 1.0 + \frac{15 - 7}{2(15) - 7 - 15}(0.5) = 1.0 + \frac{8}{8}(0.5) = 1.5\text{m/s}$$

Case 2: Taking 1.5–2.0 m/s as the modal class

$$L = 1.5, f_1 = 15, f_0 = 15, f_2 = 10, h = 0.5$$

Therefore,

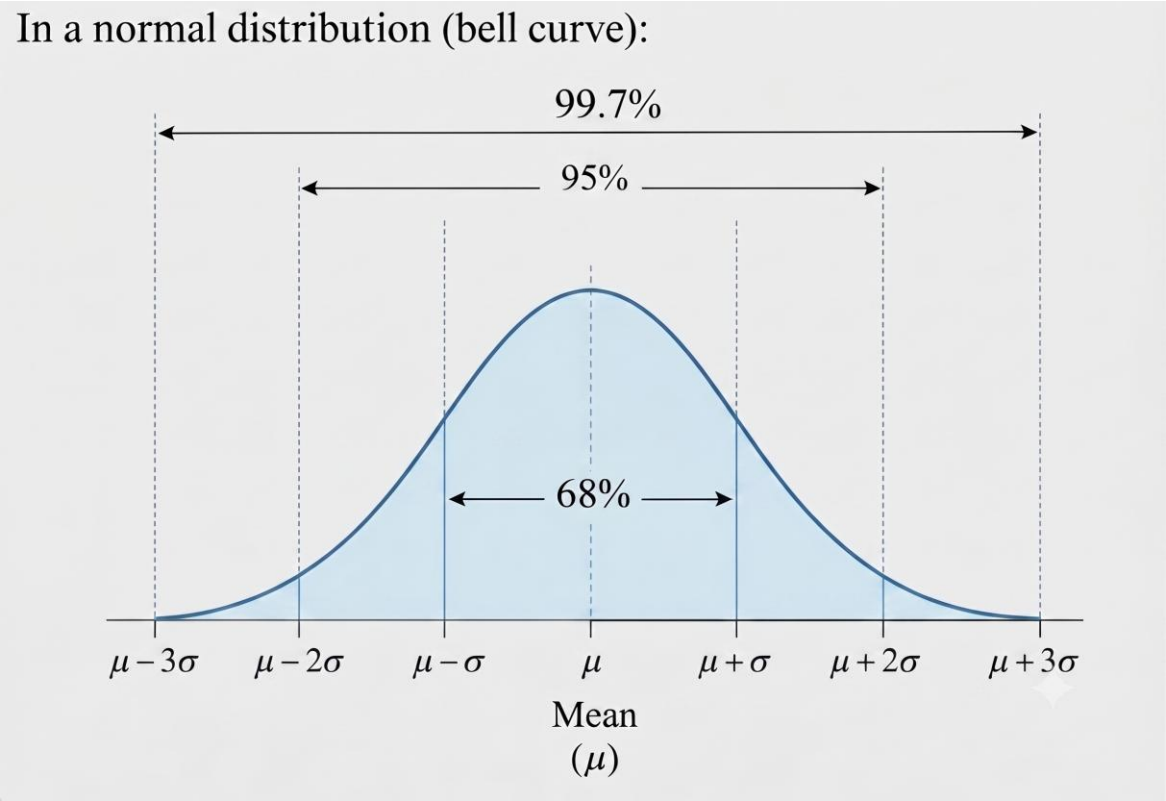
$$\text{Mode} = 1.5 + \frac{15 - 15}{2(15) - 15 - 10}(0.5)$$
$$= 1.5 \text{ m/s}$$

Types of Data Sampling

Standard Deviation (SD)

Standard deviation is a statistical measure of dispersion that quantifies the amount of variation or spread of a set of data values relative to its arithmetic mean. A low standard deviation indicates that the data points tend to be close to the mean, while a high standard deviation indicates that the data points are spread out over a wider range of values:

$$\sigma = \left[\frac{\sum_{i=1}^{n}(x_i - x_m)^2}{n-1}\right]^{1/2}$$



Variance (σ^2)

The square of the standard deviation, representing the second moment of dispersion about the mean.

Uncertainty Analysis

- Uncertainty analysis is the systematic estimation of the uncertainty in a final experimental result based on the uncertainties associated with the measured variables.

For example, if a pressure is reported as:

$$p = 100 \pm 1 \text{ kPa}$$

Where, **±1 kPa** represents the estimated uncertainty of the measurement. It indicates the range within which the true value is likely to lie.

- If a final result (R) depends on several measured quantities, such as $x_1, x_2, \ldots, x_n$

then $R = R(x_1, x_2, \ldots, x_n)$, where n = total number of measured variables

For example, if heat transfer is calculated as: $Q = \dot{m} C_p (T_{out} - T_{in})$

Then it can be written as: R = Q

And the measured variables are : $x_1 = \dot{m}$, $x_2 = C_p$, $x_3 = T_{out}$, $x_4 = T_{in}$

So $Q = Q(\dot{m}, C_p, T_{out}, T_{in})$

- Therefore, the uncertainty in Q comes from the uncertainties in $\dot{m}$, c_p, T_{out}, and T_{in}.
- Uncertainty analysis combines all these effects to determine the uncertainty of the final Q.

Uncertainty Analysis

- The Kline–McClintock uncertainty-propagation equation is used to determine how the uncertainties in several measured quantities combine to produce uncertainty in a final calculated result.

If general result is $R = R(x_1, x_2, \ldots, x_n)$

then Uncertainty calculated as: $$w_R = \left[\left(\frac{\partial R}{\partial x_1} w_1\right)^2 + \left(\frac{\partial R}{\partial x_2} w_2\right)^2 + \ldots + \left(\frac{\partial R}{\partial x_n} w_n\right)^2\right]^{1/2}$$

where $w_1, w_2, \ldots, w_n$ are the uncertainties of the individual measured variables.

- From previous example, R = Q

And the measured variables are : $x_1 = \dot{m}$, $x_2 = C_p$, $x_3 = T_{out}$, $x_4 = T_{in}$

The Kline–McClintock equation becomes $$w_Q = \left[\left(\frac{\partial Q}{\partial \dot{m}} w_{\dot{m}}\right)^2 + \left(\frac{\partial Q}{\partial C_p} w_{C_p}\right)^2 + \left(\frac{\partial Q}{\partial T_{out}} w_{T_{out}}\right)^2 + \left(\frac{\partial Q}{\partial T_{in}} w_{T_{in}}\right)^2\right]^{1/2}$$

where $w_{\dot{m}}, w_{Cp}, w_{Tout}, w_{Tin}$ are the uncertainties of the individual measured variables.

In other words, The Kline–McClintock equation determines how the uncertainties of mass flow rate, specific heat, inlet temperature, and outlet temperature propagate into the uncertainty of the calculated heat-transfer rate.

Problem

- Water flows through a heat exchanger. The heat-transfer rate is calculated from

$$Q = \dot{m} C_p (T_{out} - T_{in})$$

The measured values are

$$\dot{m} = 0.50 \pm 0.01 \text{ kg/s}$$

$$c_p = 4.18 \pm 0.04 \text{ kJ/(kg.K)}$$

$$T_{out} = 35.0 \pm 0.5^{\circ}C$$

$$T_{in} = 25.0 \pm 0.5^{\circ}C$$

Determine the heat-transfer rate and its uncertainty using the Kline–McClintock uncertainty-propagation method.

Solution

The nominal heat-transfer rate is $Q = (0.50)(4.18)(35 - 25) = 20.9$ kW

Using the Kline–McClintock equation,

$$w_Q = \left[\left(\frac{\partial Q}{\partial \dot{m}} w_{\dot{m}}\right)^2 + \left(\frac{\partial Q}{\partial C_p} w_{C_p}\right)^2 + \left(\frac{\partial Q}{\partial T_{out}} w_{T_{out}}\right)^2 + \left(\frac{\partial Q}{\partial T_{in}} w_{T_{in}}\right)^2\right]^{1/2}$$

For, $Q = \dot{m} C_p (T_{out} - T_{in})$

Partial derivates are:

$$\frac{\partial Q}{\partial \dot{m}} = C_p(T_{out} - T_{in}) = (4.18)(10) = 41.8$$

$$\frac{\partial Q}{\partial C_p} = \dot{m}(T_{out} - T_{in}) = (0.50)(10) = 5$$

$$\frac{\partial Q}{\partial T_{out}} = \dot{m} C_p = (0.50)(4.18) = 2.09$$

$$\frac{\partial Q}{\partial T_{in}} = -\dot{m} C_p = -2.09$$

The nominal heat-transfer rate is $Q = (0.50)(4.18)(35 - 25) = 20.9$ kW

Substituting the uncertainties,

$w_Q = [(41.8 \times 0.01)^2 + (5 \times 0.04)^2 + (2.09 \times 0.5)^2 + (-2.09 \times 0.5)^2]^{1/2}$

$w_Q = [0.418^2 + 0.20^2 + 1.045^2 + 1.045^2]^{1/2}$

$w_Q = (2.398)^{1/2} = 1.55$ kW

Therefore, Q $= 20.9 \pm 1.55$

The percentage uncertainty is $\frac{w_Q}{Q} \times 100 = \frac{1.55}{20.9} \times 100 = 7.4\ \%$

So the final result can also be written as Q = 20.9 ± 7.4%

Thank You